

HealthHub: A Hybrid-Intelligence Meta-App for Personalized, Sustainable Food Choices with Adaptive Nudging

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Abstract

Consumers today navigate fragmented food services (such as recipes, grocery shopping, and delivery) without unified, health/sustainability-aware guidance. Existing recommender systems often prioritize clicks over long-term personal well-being and global sustainability goals. We introduce HealthHub, a scalable meta-platform that (1) aggregates real-time data from recipes, retailers, and restaurants, (2) employs a multimodal, multi-objective AI engine (optimizing for taste, health, and sustainability) enhanced by wearable sensing, (3) leverages LLM-powered, explainable nudges, and (4) incorporates gamified behavior-change mechanics.

By integrating cross-service infrastructure with hybrid-intelligence framework, HealthHub illustrates how ubiquitous and wearable computing (via always-on wristband sensing, context-aware notifications, and on-device AI) can adapt recommendations in real time to a user's physiological and situational context. It transparently justifies each suggestion and fosters lasting shifts toward healthier, more sustainable and lower-carbon diets.

CCS Concepts

• **Applied computing** → **Health informatics**; • **Human-centered computing** → **Ubiquitous computing**.

Keywords

Ubiquitous Computing, Personalized Recommendation, Mobile Health (mHealth), LLM, Restaurant Recommendation Systems

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1 Introduction

Sustainability has emerged as an urgent global priority. The United Nations Sustainable Development Goals (SDG) call for responsible consumption and production (SDG 12), decisive climate action (SDG 13), and sustainable cities and communities (SDG 11), while also promoting healthy lifestyles (SDG 3), equitable access to nutritious

food (SDG 2), and inclusivity in resource distribution (SDG 4 & SDG 5). Yet achieving these interconnected objectives demands tools that can seamlessly guide individual behavior toward both personal well-being and environmental stewardship.

In this context and to promote healthy and more sustainable food choices, future efforts must emphasize the importance of fostering behavioral change to encourage sustainable actions, rather than solely focusing on fulfilling immediate needs or desires [5, 6]. This approach involves creating long-term strategies that prioritize education, awareness, and the development of systems that support environmentally and socially responsible choices. The generic suggestion and recommendation provided by these systems, while aligned with broad sustainability goals, often fail to resonate with individuals unless they address personal preferences, cultural contexts, and situational constraints. For example, EU's dietary shifts toward sustainable food practices would have limited success without strategies that guide individuals to actionable, relevant choices tailored to their specific needs and lifestyles.

Behavioral change initiatives are most effective when implemented through platforms where people actively engage, such as social media [12]. In the last decade, the use of social media has revolutionized daily life, influencing how people access information, make decisions, and shape their lifestyles [14]. Social media and online apps, combined with the rapid growth of e-commerce platforms, have significantly altered consumer behavior and choices [19]. In the EU, where internet penetration stands at 93.1% and approximately 60% of the population actively uses social media [9], these platforms present a powerful avenue for promoting sustainable interventions. Leveraging social media for sustainability efforts can ensure wide-reaching, real-time engagement with individuals, making it a powerful tool to drive collective behavioral change [22].

Modern food ecosystems are highly fragmented: consumers bounce between recipe platforms, grocery apps, and restaurant services, each optimizing for convenience or clicks rather than long-term health or carbon footprint reduction. This siloed experience undermines sustained dietary change, contributing to the global rise in chronic diseases (obesity, diabetes, cardiovascular conditions) and food-system greenhouse gas emissions, which currently account for over 26% of total emissions. Ubiquitous and wearable computing offer a transformative avenue to bridge this gap. Always-on wristbands and context-aware mobile interfaces can continuously sense physiological signals (e.g., heart-rate variability, stress markers), environmental cues (time, location, social context), and user interactions to deliver timely, personalized interventions.

By embedding AI and LLM-powered nudges directly into users' daily routines, such systems can adapt recommendations in real



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time, turning fleeting moments of decision (“What should I eat now?”) into opportunities for healthier, lower-carbon choices. HealthHub leverages these capabilities, integrating a scalable meta-platform for cross-service food data aggregation, an engine that balances taste, health, and sustainability, and a gamified behavior-change loop, to demonstrate how ubiquitous sensing and adaptive nudging can foster lasting shifts toward sustainable, health-promoting diets.

2 Personalization for Sustainability

2.1 Research-based efforts

Technology, particularly artificial intelligence (AI), has become a critical enabler in achieving these objectives. Among AI applications, recommender systems (RS) stand out for their ability to provide personalized suggestions tailored to user preferences and behaviors [11, 15]. Recommender systems have revolutionized decision-making across diverse domains, leveraging user data to personalize experiences and improve outcomes. In entertainment, platforms like Netflix [18] and Spotify [4] utilize advanced algorithms to suggest content, enhancing user engagement and satisfaction by catering to individual preferences. In e-business, recommendation engines drive sales by suggesting products tailored to users’ browsing and purchase histories, significantly boosting revenue and customer retention [13]. Similarly, in healthcare, recommender systems play a pivotal role by providing personalized health advice, recommending preventive measures, or suggesting treatments based on patients’ medical history and preferences, thereby supporting better health outcomes [20]. These systems have become indispensable tools, transforming the way decisions are made, ensuring efficiency, and fostering informed choices in various sectors. This personalization is essential for driving sustainable behaviors, as generic recommendations often fail to resonate with users and are likely to be ignored or rejected. RS have the potential to align individual actions with global sustainability goals by delivering actionable, context-specific recommendations. RS have demonstrated their capacity to address various aspects of the SDG (See Figure. 1):

- **Promoting Healthy Lifestyles:** Personalized healthcare [16], physical activity [10], and exercise [2] contribute to improved public health, aligning with SDG 3. Moreover, food recommendations [21] support balanced and sustainable diets, addressing SDG 3, SDG 2 and SDG 12. While international initiatives, such as the EU-funded *SUSFOOD2*, Horizon Europe projects, and academic efforts from institutions like *University of California Irvine* [23], *Technical University of Munich* [3], and *University of Bergen* [17] have explored machine learning and behavioral economics in food domain choice and focused on health or sustainability through digital interventions, they often operate in silos, addressing health, sustainability, or technical modeling separately. Also, these efforts did not incorporate comprehensive *nudging strategies* aimed at altering users’ lifestyles and eating habits toward more sustainable/healthier choices, which is critical to long-term behavioral change.
- **Fostering Eco-Friendly Travel Behaviors:** Sustainable travel RS [1] mitigate overtourism and promote environmentally friendly travel options, supporting SDG 11 (Sustainable Cities and Communities) and SDG 13 (Climate Action).

- **Sustainable Design of Recommendation System:** Beyond recommending sustainable content, their eco-efficiency is critical, focusing on reducing energy consumption, carbon emissions, and resource usage by developing eco-efficient models, aligning with SGD 12 (Responsible Consumption and Production) and SGD 13 (Climate Action), by promoting resource efficiency and reducing carbon footprints in computational processes. Moreover, by developing fairness-aware RS with bias mitigation strategies, they promote equitable access to resources, thereby contributing to SDG 4 (Quality Education), SDG 5 (Gender Equality), SDG 8 (Decent Work and Economic Growth), and SDG 10 (Reduced Inequalities).

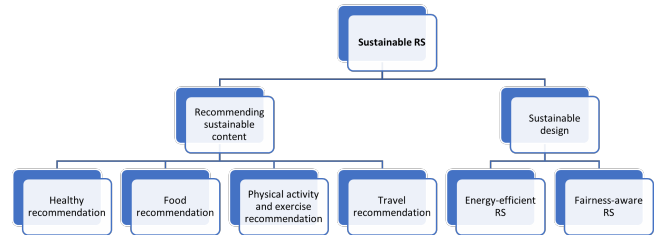


Figure 1: Different capacity of RS in advancing sustainability

2.2 Existing Apps in the Food & Sustainability

Table 1 categorizes existing food-related apps into seven domains (ranging from on-demand delivery and grocery ordering to personalized nutrition and carbon footprint estimation) listing representative services in each category. For each, it concisely summarizes core functionality (e.g., “On-demand restaurant delivery” or “GHG estimation for meals”) and highlights key gaps (such as lack of healthy or low-carbon recommendations, absence of personalized nudges, or limited real-time integration). This overview makes it clear that while many apps address single aspects of sustainability or health, none combine dynamic, user-centered recommendation with both environmental and dietary goals.

HealthHub fills a critical gap by unifying the strengths of waste-reduction apps, dietary trackers, carbon calculators, and gamified behavior tools into a single, adaptive platform, pioneering a new paradigm of personalized, sustainability-oriented recommendations in the realm of ubiquitous and wearable computing.

3 System Design & Novelty

HealthHub’s architecture comprises six interconnected components (Data Integration, Dynamic Profiling, Analysis & Scoring, LLM-Powered Nudging, Gamification, and User Feedback & Refinement) each designed to leverage ubiquitous and wearable computing for real-time, personalized, sustainability-oriented guidance.

Data Integration Layer

HealthHub continuously aggregates content from external food platforms (recipe sites, grocery store inventories, and restaurant menus) via lightweight APIs and web hooks. Concurrently, it ingests user-provided preferences (dietary restrictions, allergy information,

Table 1: Concise Comparison of Existing Food-Related Sustainability and Health Apps

Category	Example Apps	Key Features	Limitations
Convenience-Based Delivery	Wolt / Foodora	On-demand restaurant delivery	No healthy/low-carbon recommendations; no behavioral nudges
Grocery Ordering & Retail	Instacart / Walmart Grocery	Online grocery/surplus ordering	No personalized health/sustainability recommendations; no timely nudges
Food Waste Reduction & Surplus Redistribution	Too Good To Go / Olio / ResQ Club / Fiksuruoka	Surplus food discounts	Transactional only; no long-term dietary guidance
Nutrigenomics & Personalized Nutrition	Habit / DNAfit	Genetic/blood-based diet plans	High cost; lab-based; no real-time integration
Nutrition & Fitness Tracking	MyFitnessPal / Lifesum / Yazio	Manual meal/exercise logging	No service integration; no carbon scoring
Carbon Footprint & Sustainability Estimation	FoodPrint (ReFed) / Eaternity	GHG estimation for meals	Static input; no dynamic or contextual recommendations
Sustainable Travel (Adjacent Domain)	GreenTraveler / JouleBug	Eco-friendly travel suggestions	No food or health integration

long-term health goals) alongside live biometric signals from wearable devices (heart rate variability, step count, sleep patterns, stress markers). By unifying these heterogeneous sources into a single meta-platform, HealthHub maintains an up-to-date catalog of food options enriched with nutritional profiles and environmental meta-data (e.g., estimated carbon footprints, water usage).

Dynamic Profiling Engine

The Dynamic Profiling Engine continuously updates each user’s profile by fusing two primary inputs:

- **User Preferences:** Incorporates self-reported health data (e.g., medical conditions, caloric targets) and dietary restrictions (e.g., vegan, gluten-free), along with historical behavior such as past meal selections and explicit ratings.
- **Wearable Sensing Insights:** Captures real time biometric signals (e.g., heart rate, activity levels and sleep quality) using wrist-worn/smartwatch devices. These signals flag moments of high receptivity (e.g., post-exercise) or physiological need (e.g., elevated heart rate indicating hunger).

Analysis & Scoring Module

Using the combined catalog and dynamic profile, this module evaluates candidate food items across three objectives:

- **Taste Alignment:** A collaborative filtering component infers similarity to previously enjoyed dishes, refined by image-based recognition (food photos) and sentiment analysis of textual reviews.
- **Health Optimization:** A nutrient-profiling subcomponent computes macro-/micronutrient compliance relative to individual health targets (e.g., caloric budgets, sodium limits).
- **Sustainability Impact:** An environmental scoring subcomponent estimates carbon emissions and resource footprints at the ingredient level to assign a “sustainability index.”

These three scores feed into a multi-objective optimization layer that ranks items based on personalized, long-term benefit rather than short-term clicks.

LLM-Powered Nudging System

Once top-ranked items emerge from Analysis & Scoring, the Nudging System generates concise, transparent, natural-language explanations on-device (e.g., “This vegan curry contains 30% more fiber than your usual stir-fry and reduces CO₂ emissions by 0.8 kg”).

Leveraging on-device large language models ensures minimal latency and preserves user privacy. Notifications are delivered to the user’s wearable or mobile interface precisely when context flags (derived from wearable signals and contextual cues) indicate high receptivity, maximizing the impact of each nudge.

Gamification Module

To reinforce healthy/sustainable choices, HealthHub incorporates a dedicated Gamification Module that issues points, badges, and challenges. For example, selecting a top-ranked, low-carbon recipe earns “green points,” while completing an “eat five plant-based meals this week” challenge unlocks a new achievement badge. Progress dashboards (accessible on both smartwatch and smartphone) visualize cumulative health metrics (e.g., protein intake, step consistency) alongside environmental savings (e.g., total CO₂ averted). By tying rewards to both behavioral and sustainability outcomes, HealthHub leverages gamified incentives to drive long-term engagement.

User Feedback & Refinement Loop

After receiving recommendations through nudges and gamification, users can provide explicit feedback: a thumbs-up/down, a comment on taste, or a manual override to log a different choice. This feedback is fed back into core personalization components (i.e., Dynamic Profiling, Analysis & Scoring, LLM nudging parameters, and gamification) enabling continuous refinement. For instance, if a user repeatedly rejects certain low-carbon items, the system adjusts the sustainability weight for that individual and tailors future suggestions accordingly. Over time, this closed-loop mechanism ensures that HealthHub’s recommendations become increasingly aligned with both evolving preferences and real-world constraints.

Novelty Highlights

- **Comprehensive Meta-Layer Integration:** Unlike single-purpose apps (e.g., grocery-only platforms or food-waste tools), HealthHub aggregates recipes, retail inventories, and menus in real time without requiring vendor cooperation, creating a unified discovery layer.
- **Hybrid-Intelligence Dynamic Profiling:** By fusing on-device wearable data (physiological signals such as heart rate, activity, sleep) with explicit user preferences (health data, dietary restrictions), HealthHub’s profiling adapts to both stable and temporary user needs, outperforming traditional RS reliant on static histories.

- **Multi-Objective, Context-Aware Optimization:** HealthHub simultaneously balances taste satisfaction, nutritional quality, and environmental impact through reinforcement learning and contextual bandits, whereas most existing systems optimize only one or two objectives.
- **Transparent, On-Device Nudges:** Leveraging on-device LLMs ensures low-latency, privacy-preserving explanations, delivered at precisely the right moments as indicated by wearable sensor inputs.
- **Dedicated Gamification & Continuous Refinement:** By separating gamification into its own module and closing the feedback loop, HealthHub continuously learns from user interactions, enabling a self-tuning system that remains responsive to evolving behavior, a capability absent in most current health or sustainability recommenders.

This unified, real-time, hybrid-intelligence approach positions HealthHub at the forefront of ubiquitous and wearable computing for sustainable, health-oriented food behavior change.

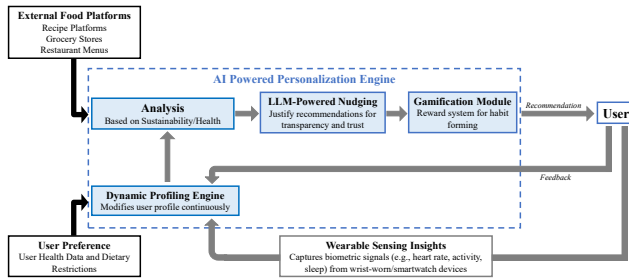


Figure 2: Flow diagram of HealthHub’s architecture

4 Conclusion & Future Plan

HealthHub addresses the pressing need for an integrated, sustainability-oriented food personalization platform by leveraging ubiquitous and wearable computing. As consumers navigate fragmented recipe platforms, grocery apps, and restaurant delivery services, they lack seamless guidance that balances personal health goals with environmental responsibility. Our proposed meta, platform unifies diverse data sources, recipes, grocery inventories, and menus, while dynamically incorporating user preferences (health data, dietary restrictions) and real-time biometric signals (heart rate, activity, sleep patterns) from wrist-worn/smartwatch devices. By combining a hybrid-intelligence profiling engine, multi-objective scoring (taste, health, sustainability), on-device LLM-powered nudges, and gamified behavior-change mechanics, HealthHub offers a novel paradigm for personalized, context-aware interventions that can foster healthier, lower-carbon dietary habits.

Feasibility and Necessity

Implementing HealthHub builds on several existing technologies and proven research insights, making it both feasible and timely:

- **API-Based Data Aggregation:** Most recipe websites, grocery services, and restaurant platforms provide RESTful APIs

or structured web data, allowing us to pull nutritional and menu information without requiring vendor integrations.

- **Wearable Sensing Frameworks:** Modern smartwatches and fitness bands routinely offer SDKs for heart rate, step count, and sleep-tracking data.
- **On-Device LLM Inference:** Recent advances in optimized language-model compression (quantization, pruning) allow for lightweight LLM inference on mobile and embedded hardware. This capability lets us generate transparent, natural-language explanations without relying on cloud servers, reducing latency and safeguarding sensitive health data.
- **Behavioral Science Foundations:** Gamification elements (points, badges, challenges) have demonstrated efficacy in domains such as physical activity promotion and nutrition counseling. Embedding these into a wearable-driven loop aligns with proven habit-formation techniques, increasing the likelihood of sustained user engagement.
- **Context-aware nudging:** Previous studies [7, 8] indicate that context-aware nudging (timed by physiological or situational cues) can significantly boost acceptance rates compared to static notifications. By integrating multi-objective recommendation with on-device LLM explanations and real-time wearable triggers, it capitalizes on these findings to offer an end-to-end solution that no current app provides.

To move from concept to prototype, our next steps include finalizing the system architecture, defining module interfaces, selecting APIs for initial integration (e.g., Allrecipes, local grocery SDK), and outlining data schemas for unified item metadata (nutrition, carbon footprint). We will then develop a feasibility prototype that demonstrates core functionality: on-device wearable data capture, basic multi-objective scoring (calorie vs. carbon footprint trade-off), and static LLM explanations for a small set of example recipes.

Following the prototype, we plan to conduct focus groups with stakeholders (dietitians, public health officials) and potential users (health-conscious individuals, sustainability advocates) to refine feature priorities and user interface preferences. These insights will inform the design of a small-scale, 10-participant field trial, including recruitment criteria, data-collection metrics (recommendation uptake, engagement duration, perceived transparency), and preliminary hypotheses on behavior change and sustainability outcomes, with the goal of laying the groundwork for iterative system improvements and larger longitudinal studies. By taking these steps, HealthHub will transition from an innovative concept to a validated, community-informed platform that can reshape how individuals make food-related decisions to support both personal well-being and global sustainability goals.

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